

Machine Learning Reviewer for SA1

CS0075 Machine Learning Algorithm, SA1. Covers Module 1 and Module 2 up to Linear Regression (OLS). Built from the slides, including the picture slides, plus the verified formative questions. The blue boxes are memory aids I made (they are not from the slides) and the red boxes are the traps.

How To Use This Reviewer

This reviewer stops at **Linear Regression (OLS)**, which is where SA1 stops. The separate midterm reviewer adds everything after it (Ridge, Lasso, Logistic Regression, Naive Bayes, decision trees, random forest, gradient boosting, and SVM). Parts: **(1) core concepts, (2) evaluation and complexity, (3) k-NN and linear models, (4) traps**. The cram sheet is the last page.

Part 1. Core Concepts and Definitions

What Machine Learning Is

- Machine learning is about **extracting knowledge from data**. It sits at the intersection of **statistics, artificial intelligence, and computer science**, and is also known as **predictive analytics** or **statistical learning**.
- Arthur Samuel, 1959**: giving computers the ability to **learn without being explicitly programmed**.
- ML is a discipline of AI that lets machines **automatically learn from data and past experiences**, identify patterns, and make **predictions** with **minimal human intervention**.

The Formal Definition (E, T, P)

A program learns from **experience E** with respect to a class of **tasks T** and **performance measure P** if its performance at T, as measured by P, **improves with experience E**.

Memory Aid

Experience, Task, Performance. Every Try Pays off: more experience, better score.

AI, ML, Deep Learning, and Data Science

- The slide diagrams nest them: **Artificial Intelligence contains Machine Learning, which contains Deep Learning**.
- Data Science** is collection, preparation, and analysis of data, leveraging AI/ML, research, industry expertise, and statistics to make business decisions.
- AI** is technology for machines to understand, interpret, learn, and make "intelligent" decisions, and includes ML among many other fields.

- **ML** is algorithms that help machines improve through supervised, unsupervised, and reinforcement learning. It is a **subset of AI** and a **Data Science tool**.

Rule-Based Algorithm vs Machine Learning

	Traditional Rule-Based Algorithm	Machine Learning
What drives it	A condition (true goes to Code 1, false goes to Code 2)	Training data goes into the ML algorithm, which produces a model
How rules arise	Explicit programming ; the rules can be manually specified	Rules are automatically learned by machines
Best when	Rules are simple	Decision-making rules are complex or difficult to describe ; samples are used for training
Prediction	Not applicable	New data goes into the model, which outputs a prediction

Memory Aid

Rule-based: **Rules** in, **Answers** out. ML flips it: **Data** and answers in, the **Rules** out.
 Fill-in-the-blank: "Rule-based is **Condition**. Machine learning is **Data**."

The scenario chart plots **rule complexity** (simple to complex) against **scale of the problem** (small to large):

	Small Scale	Large Scale
Complex rules	Manual rules	Machine learning algorithms
Simple rules	Simple problems	Rule-based algorithms

Why Machine Learning

- Computers operate **autonomously without explicit programming**; fed new data they learn, grow, and adapt.
- Performance **adaptively improves as the number of available samples increases**.
- Machines work **24/7**, need no breaks, cost much less to maintain than a human in the long run, and suit tasks that are **routine, repetitive, or tedious**.
- Automation also **mitigates risks caused by fatigue or inattention**. Slide examples: self-driving cars and robotic arms on production lines.
- Application areas on the slides: healthcare, automobile, transportation, manufacturing, e-commerce, insurance; medical diagnosis, image and speech recognition, fraud detection, recommendations, spam filtering, self-driving cars, traffic prediction, stock market trading.

Types of Machine Learning and the Workflow

The types-of-ML picture shows **four**: **Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning**. The "How does ML work" text lists three under Learning (supervised, unsupervised, reinforcement).

Phase	Steps
Phase 1: Learning	Training data, preprocessing (clean data, format data), learning, testing (measure performance, test algorithm)
Phase 2: Prediction	New data goes into the trained model , which outputs predicted data

In the picture, the training data trains the ML algorithm, then **accuracy** is checked: **unacceptable** sends you back to train again, **acceptable** gives a **successful model**.

The Four-Stage ML Workflow (DataCamp Slide)

Stage	Steps
1. Project setup	Understand the business goals; choose the solution to your problem
2. Data preparation	Data collection, data cleaning, feature engineering, split the data
3. Modeling	Hyperparameter tuning, train your models, make predictions, assess model performance (a repeating cycle)
4. Deployment	Deploy the model, monitor model performance, improve your model

Memory Aid

Set up, Prepare, Model, Deploy: Some People Make Dinner.

Programming Languages and Python Tools

- The top programming languages for ML, in rank order: **1 Python, 2 R, 3 Java, 4 Julia, 5 Scala, 6 C++, 7 JavaScript, 8 Lisp, 9 Haskell, 10 Go**.
- **Why Python:** easy-to-read syntax, extensive libraries and frameworks, strong community support, flexibility, compatibility with other languages, scalability and performance.

Name	What The Slide Says
Scikit-learn	The most prominent Python library for ML
NumPy	A fundamental package for scientific computing (mathematical functions)
SciPy	A collection of functions for scientific computing
Matplotlib	The primary scientific plotting library (data visualizations)
Seaborn	Data visualizations
Pandas	A library for data wrangling and analysis (data analysis and manipulation)
TensorFlow	Machine learning
Keras	Deep learning

PyTorch	Machine learning
Scrapy	Web crawling
SQLModel	Interacts with SQL databases
Anaconda	A Python distribution for large-scale data processing, predictive analysis, and scientific computing
Jupyter Notebook	An interactive environment for running code in the browser

Watch Out

Formative Q5 ("most popular libraries, choose all") keyed **PANDAS, SCIPY, NUMPY**. Anaconda is a **distribution** and Jupyter is an **interactive environment**, so they are not libraries. SQL is a query language.

Data Preprocessing

Real data has three quality problems: **incompleteness** (missing values or attributes), **noise** (incorrect records or exceptions), and **inconsistency** (inconsistent records). "**Without good data, there is no good model.**" Preprocessing (also called data preparation) is cleaning, altering, and reorganizing raw data before analysis.

A data scientist's time (pie chart): **60 percent data cleaning and organization**, 19 percent data set collection, 9 percent data mining, 4 percent algorithm improvements, 3 percent training set creation, 5 percent other.

Errors The Slide Marks In A Messy Table

Error	Example Meaning
Missing value	A blank cell
Invalid value	A value outside what is allowed (for example a letter in a gender column that expects M or F)
Invalid duplicate item	The same Id appears twice
Value in another column	A value that should be in a different column
Misspelling	For example "Ytali" for Italy
Incorrect format	A date written in a different format
Attribute dependency	Columns whose values contradict each other

Key Steps in Order

Data profiling, data cleansing, data reduction, data transformation, data enrichment, data validation.

Category	Techniques
Data Cleansing	Identify and sort out missing data , reduce noisy data , identify and remove duplicates
Feature Engineering	Feature scaling or normalization, data reduction, discretization, feature encoding

Outliers

An **outlier** is an observation that seems to be **distant from other observations** or, more specifically, one observation that **follows a different logic or generative process** than the other observations. **A single outlier can throw off a regression estimate** (the regression line gets pulled toward it).

Exploratory Data Analysis (EDA)

EDA understands the main characteristics of the data, spots patterns and **anomalies**, tests a hypothesis, or checks assumptions, often with visualization.

Activity	What It Does
Visualization	Plots and charts of distributions and relationships. Bar plots and histograms visualize the count of values
Summary Statistics	Mean, median, variance, standard deviation
Outlier Detection	Identifying unusual data points
Correlation Analysis	Examining relationships between variables
Hypothesis Testing	Testing initial assumptions about the data

Memory Aid

The point is the **outlier**; the process is **outlier detection**. "Identifies unusual data points" means the process.

The Titanic Example Used In The Slides

- **Titanic Survival Data Set:** 1309 passengers and whether they survived; the goal is to **predict survival from the attributes**. Tabular data can have **different data types in one table**.
- **Types:** **Survived** is integer binary (the **label**); **Pclass** is integer categorical or ordinal; **Name** and **Ticket** are strings; **Sex** and **Embarked** are string categorical; **Age** and **Fare** are continuous; **SibSp** and **Parch** are non-negative integers.
- **Feature selection:** dropping columns that are not useful (for example Name, Ticket, Cabin).
- **Missing values:** Age had **177** missing values and Embarked had **2**. Two options: **remove rows** with missing values, or **imputation** (filling in missing values by estimating them).
- **Mean imputation** (for a numeric column) and **most-frequent imputation** (mode, for a categorical column). Other imputers: median, iterative, kNN.
- **Column transformation** before training: **standard scaling** for numerical data (Age, Fare), **ordinal encoding** for categorical data (Pclass, Sex, Embarked); other features are retained.

Discrete vs Continuous Data

Type	Meaning	Example
Discrete	Countable in whole units; you cannot have a fraction of it	Number of students in a class
Continuous	Lies on an unbroken scale	Height, weight,

		temperature, Age, Fare
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Memory Aid

Discrete = Digits you can count. Continuous = Can slide anywhere on the scale.

Model Deployment

Deployment means taking a trained model and making it available in real applications. The stages are **Training, Validation, Deployment, Monitoring** (models can **drift** as real-world data evolves). Deployment means defining real-time data handling, storage, predictions, APIs and tools, hardware (cloud or on-prem), and a pipeline for continuous training. Best practices: **(1) choose the right infrastructure, (2) effective versioning and tracking, (3) robust testing and validation, (4) implement monitoring and alerting.**

Ethics and Privacy in Data Science

Five Principles	Meaning
Ownership	An individual owns their personal information; collecting it without consent is unethical
Transparency	Data subjects have a right to know how it is collected, stored, and used
Privacy	Protect personally identifiable information (PII): full name, birthdate, phone number, bank account number, street address
Intention	Ask why you need the data and what you will gain; malicious intent is unethical
Outcomes	Even good intentions can cause harm; unintended harm to a group is disparate impact

Memory Aid

Own it, Tell them, Protect it, Intend well, check Outcomes: Our Team Protects Individuals' Ownership.

Topic	What The Slides Say
Three new rules of data	Trust over transactions (consent); Insight over identity (re-think transferring PII); Flows over silos (teams share the flow of insights)
Regulations shown	GDPR (General Data Protection Regulation) and the National Privacy Commission (Philippines)
Data subject rights	Right to be informed , to damages , to access , to erasure or blocking , to file a complaint , to object , to rectify , and to data portability
Data governance	A set of principles and processes for data collection, management, and use, so data is accurate, consistent, and available while protecting privacy and security. A framework describes how to do what governance says the organization

	needs to do
Pillars of data governance	Ownership and accountability, data quality, data protection and safety, data use and availability, data management (built on people, processes, and technology)
10 questions before AI in public decisions	Objective, use, impacts, assumptions, data, inputs, mitigation, ethics, oversight, evaluation

Part 2. Evaluation, Generalization and Complexity

Supervised Learning Basics

- **Labeled training data** has both the **input** (data features) and the **correct output** (the label or goal). Each example is a pair: an input object (feature vector) and a desired output value.
- The primary objective is to **map the input variable to the output variable**: $y = f(x)$. In the slide example, weather, temperature, and wind speed are the input x and "Enjoy Sports" (yes or no) is the output y .
- Two phases: **training and validation**, followed by **prediction**.

	Classification	Regression
Target	Categorical variable	Continuous variable
Predicts	One of the possible class labels	Consecutive numbers (real numbers) , per the slide
Examples	Spam or not spam, COVID positive or negative on a chest X-ray, handwritten digit recognition	Stock price forecast, weather prediction, annual income forecast, population forecast
Subtypes	Binary (two classes) and multiple (three or more)	Simple linear, multivariate
Algorithms (slide)	Random Forest, Decision Tree, Logistic Regression, Support Vector Machine	Simple Linear Regression, Multivariate Regression, Decision Tree, Lasso Regression

Watch Out

Formative Q17. The slide says regression "**predicts consecutive numbers (real numbers)**". The graded formative key marked "Regression predicts consecutive numbers" as **False** (regression predicts **continuous real values**, not a counting sequence like 1, 2, 3). Study rule: **regression = continuous real numbers**. If a true or false item says "consecutive numbers" with no mention of real numbers, go with the **formative key (False)**, because your teacher wrote both the quiz and the exam. If the item says "real numbers", it is True.

Generalization, Overfitting, Underfitting

Term	What The Slides Say	Train	Test
Generalization	The model makes accurate predictions on new ,	Good	Good

	unseen data with the same characteristics as the training set. We want to generalize as accurately as possible .		
Overfitting	The model fits the particularities of the training set (its noise and outliers) too closely; it has memorized the training data rather than learned the pattern.	Very good	Poor
Underfitting	The model is too simple to capture the variability in the data and does badly even on the training set .	Poor	Poor

Memory Aid

Under = Underbuilt (bad everywhere). **Over = Overthinks** (aces training, flops on new data).

Model Complexity and the Complexity Curve

- The curve plots **accuracy** against **model complexity**. **Training accuracy keeps rising** as complexity rises. **Generalization (test) accuracy rises, peaks at the sweet spot, then falls**.
- **Left side = underfitting** (too simple). **Right side = overfitting** (too complex). The **sweet spot** in between gives the **best generalization**.
- The more complex the model, the better it predicts the **training** data. Too complex, and it focuses on each individual point and **does not generalize**.
- The larger the **variety** of data points, the more complex a model you can use without overfitting. **Duplicating the same points or collecting very similar data does not help**.
"Never underestimate the power of **more data**."

Complexity	Fit Problem	Error Type (Slide Definition)
Too low (too simple)	Underfitting	Bias = error from wrong or too-simple assumptions in the algorithm
Too high (too flexible)	Overfitting	Variance = error from sensitivity to noise and fluctuations in the training data

The **dartboard** picture: **low bias and low variance = good model** (darts tightly clustered on the bullseye). High bias means the cluster is off the bullseye; high variance means the darts are spread out.

Watch Out

"Low complexity is overfitting" is **False**. Low complexity is **underfitting**.

The boat example: a rule that is 100 percent accurate on 12 rows is not trustworthy. If 10,000 more rows obeyed the same rule, you would believe it. More data supports a more complex model.

R-Squared (R², Coefficient of Determination)

$R^2 = 1 - [\text{sum of } (y - \hat{y})^2] / [\text{sum of } (y - \bar{y})^2]$, where y is the target value, $\bar{y}$ the average value, and $\hat{y}$ the predicted value.

Situation	R ²
Perfect prediction (prediction equals target, numerator equals zero)	1
Predicting the average (numerator equals denominator): a constant model that predicts the mean of y_{train}	0
Worse than predicting the average	Negative

Memory Aid

R² is a **report card against guessing the average**: 1 is perfect, 0 is no better than the average, below 0 is worse than the average.

Watch Out

The slide also says R² "yields a score between 0 and 1", but the next slide says it **can be negative**. "Predicting worse than the average can give a negative number" is **True** (formative Q11). For regressors, scikit-learn's **score()** returns **R²**; for classifiers it returns **accuracy**.

Part 3. The Algorithms

3.1 k-Nearest Neighbors (k-NN)

- Arguably the **simplest** ML algorithm. **Building the model is only storing the training dataset**.
- To predict, it finds the **closest training points** (the nearest neighbors).
- **Classification**: with $k = 1$, the prediction is the label of the single nearest point. With $k > 1$ it uses **voting**: the **majority class** among the k neighbors wins. Works for any number of classes.
- **Regression**: with $k = 1$, the prediction is the target of the nearest neighbor. With $k > 1$ the prediction is the **average (mean)** of the neighbors.
- In scikit-learn: **KNeighborsClassifier** and **KNeighborsRegressor**; train with **fit**, predict with **predict**, evaluate with **score**.

	Small k (1 neighbor)	Large k (9 neighbors)
Decision boundary	Jagged; predictions go through every point (unsteady)	Smoother ; does not fit the training data as well
Model complexity	More complex	Less complex
Bias and variance	Bias goes down , variance goes up	Bias goes up , variance goes down
Risk	Overfitting	Underfitting

On the **accuracy against n_neighbors** plot, **training accuracy falls** as n_neighbors grows, test accuracy peaks around 6, and **the best performance is somewhere in the middle** (the optimal point). The left side is overfitting and the right side is underfitting.

Memory Aid

Small k = Small crowd = Super complex = Overfit. Big k = Big crowd = Blurry average = Underfit.

Watch Out

"A small k makes the model less complex" is **False** (formative Q9). "In k-NN, low model complexity is overfitting" is **False** (Q10): low complexity is underfitting.

Important Parameters	Strengths	Weaknesses
Number of neighbors and the distance measure (Euclidean by default) . Three or five neighbors usually works well, but adjust it.	Easy to understand; works well without special adjustments ; suitable as a first model	Slow prediction when features or samples are many, so preprocessing matters; does not work well with sparse datasets

3.2 Linear Models

- Linear models make a prediction using a **linear function of the input features**: a **best-fit line**.
- They assume a **linear relationship** between the outcome and each predictor. Examples: boiling point vs altitude, advertising spend vs revenue, fertilizer vs crop yield, athlete performance vs training.
- Types: **Linear Regression** (regression, numeric output) and **Logistic Regression** (classification).

The Prediction Formula

$$\hat{y} = w[0]*x[0] + w[1]*x[1] + \dots + w[p]*x[p] + b$$

Symbol	Meaning
x[0] to x[p]	The features (p + 1 of them)
w[0] to w[p]	The weights or coefficients (the slopes)
b	The intercept , also called the offset or bias
y-hat	The model's prediction

Memory Aid

w = Weight = slope. b = Bias = intercept = offset. All three of intercept, offset, and bias point at **b**.

In scikit-learn the weights are stored in **coef_** and the intercept in **intercept_**. "The offset parameter is also called the intercept" is **True** (formative Q12).

3.3 Linear Regression (Ordinary Least Squares)

- **Linear regression = ordinary least squares (OLS)**: the simplest and most classic linear method for regression.
- It finds **w** and **b** that **minimize the mean squared error** between predictions and the true targets on the **training set**.
- The **mean squared error** is the sum of squared differences between the predictions and the true values (as the slide words it).

Example	Train R2 vs Test R2	Diagnosis
Wave dataset (one feature)	About 0.66 , and the train and test scores are very close	Likely underfitting , not overfitting
Extended Boston Housing (506 samples, 104 or 105 features)	Very accurate on train , much worse on test	Overfitting . Too many features inflate the weights. Try a model that controls complexity : Ridge

The extended Boston data is the **13 original features plus the 91 two-feature products** (the slide's shape is (506, 104); the slide text elsewhere says 105 features). Including derived features like these is **feature engineering**.

Memory Aid

Close scores, mediocre = underfit. Big gap, high train = overfit.

SA1 stops here. The midterm reviewer continues with Ridge, Lasso, Logistic Regression, Naive Bayes, decision trees, random forests, gradient boosting, and SVM.

Part 4. High-Yield Traps

These are the twists the formatives used, plus the slide details most likely to be tested. Most are a true idea with **one word flipped**.

Trap	Wrong Idea	Right Idea
Discrete vs continuous	Number of students is continuous	Discrete : countable whole units
Consecutive vs continuous	Regression predicts counting numbers	Regression = continuous real numbers . Formative key: "consecutive numbers" is False
Small k	Small k makes the model less complex	Small k = more complex = overfitting. False .
Low complexity	Low complexity is overfitting	Low complexity is underfitting . False .
Rule-based vs ML	ML runs on conditions	Rule-based = condition ; ML = data
Types of ML	There are three types	The picture shows four : supervised, unsupervised, semi-supervised ,

		reinforcement
Libraries	Anaconda or Jupyter is a library	Libraries: pandas, SciPy, NumPy , etc. Anaconda = distribution; Jupyter = environment
Outlier word	Answering the process for the point	Point = outlier ; process = outlier detection
Missing values	Only one way to handle them	Remove rows or impute (mean for numeric, most frequent for categorical)
Python's rank	Java is the top ML language	Python is 1 , R is 2, Java is 3
R2 sign	R2 can never be negative	Worse than the mean gives negative R2. True.
Offset	Offset is not the intercept	Offset = intercept = bias (b). True.
score()	score() is always accuracy	Classifier = accuracy; regressor = R2

Cram Sheet

Everything on one page.

- **ML** = extracting knowledge from data; learns from **data / experience** to predict with minimal human intervention. **Samuel, 1959. AI contains ML contains Deep Learning.**
- **E, T, P:** performance at task T, measured by P, improves with experience E.
- **Rule-based = condition**, manually specified rules; **ML = data**, rules learned automatically.
- **Four types** (picture): supervised, unsupervised, **semi-supervised**, reinforcement. Workflow: **project setup, data preparation, modeling, deployment.**
- Python is **#1** (then R, Java). Libraries: **pandas, NumPy, SciPy, Matplotlib, Seaborn**, scikit-learn, TensorFlow, Keras, PyTorch.
- Data problems: **incompleteness, noise, inconsistency**. Preprocessing steps: **profiling, cleansing, reduction, transformation, enrichment, validation**. Data scientists spend **60 percent** on cleaning.
- **Outlier** = far from the rest; **outlier detection** = EDA process. Missing values: **remove rows or impute** (mean, most frequent).
- **Discrete** = countable. **Continuous** = unbroken scale. **Classification** = categorical target. **Regression** = continuous target.
- **Generalization** = good on unseen data. **Underfitting** = too simple, **high bias**. **Overfitting** = memorized training, **high variance**. **Low bias + low variance = good model.**
- **Small k** = complex = overfit. **Large k** = simple = underfit. k-NN: store data, vote (class) or average (regression), Euclidean by default.
- **R2** = $1 - SS_{\text{res}} / SS_{\text{tot}}$: 1 perfect, 0 = predicts the mean, **negative = worse than the mean**.
- **y-hat** = $\mathbf{w} \cdot \mathbf{x} + \mathbf{b}$; **b** = intercept = offset = bias. **OLS** minimizes mean squared error on the training set.
- OLS: small gap and mediocre score = underfit; huge train-test gap (many features) = overfit.